AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 3 · STUDENT LESSON

Bias terms and decision boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Compute the score. Draw the boundary. Check equality.

Explain how a declared bias shifts a toy boundary while weights retain its slope.

Keep one habit: Complete formula → labelled original/shifted graph → pictured two-input trace → equality and unit probes → new independent model.

MAIN LESSON · SECTION 1

Where does the score change class?

> Keep the formula,axes and equality rule visible.

Learning goals

I can calculate a weighted score with a **bias term**.

I can locate and test a two-feature **decision boundary**.

I can explain how changing the bias shifts a boundary without **training**.

Retrieve signed contributions

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite calculation

- For w = [2, -1] and x = [3, 5], what are the contributions and dot product?

> **My starting explanation**

End goal

By the end, I can explain **bias term**s and decision boundaries using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Calculate a weighted score with a **bias term**

Locate and test a two-feature **decision boundary**

Explain how changing the bias shifts a boundary without **training**

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Add one bias after the weighted sum

> The declared comparison turns an affine score into a toy class.

Compute the offset and name the equality rule

Use the fictional parcel schema [width in cm, height in cm]. Our model computes z = 2 times width - height - 3. The weights are [2, -1]; the bias b is -3. Compute the weighted sum first, then add the bias once.

The result is an affine score: a weighted sum plus a constant offset.

Now declare a classification rule: predict class 1 when z is at least 0, and class 0 when z is below 0. Equality belongs to class 1. These are invented categories for arithmetic practice, not a real parcel decision.

The weights and bias are supplied, so this is a forward calculation, not **training**.

Complete score and class examples

Input [width, height]	Score z	Prediction
[4, 2]	8 - 2 - 3 = 3	1
[2, 4]	4 - 4 - 3 = -3	0
[3, 3]	6 - 3 - 3 = 0	1 at equality

Distinguish mathematical bias from unfair treatment

A bias is a mathematical parameter. It is not the same use of the word as unfair treatment; numerical offset alone does not establish whether a system is fair.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Derive the line,then check both sides

> Plotting a line is incomplete without the axis labels and equality rule.

Complete algebra and side checks

From score to plotted boundary	Deciding step
Start with the declared score	$z=2\times\text{width}-\text{height}-3$
Locate score zero	$0=2\times\text{width}-\text{height}-3$
Isolate the vertical coordinate	$\text{height}=2\times\text{width}-3$
Plot complete boundary points	(2, 1), (3, 3), (4, 5), axes in cm
Check below and above at width 4	height 4:$z1$, class 1; height 6:$z-1$, class 0
Check equality explicitly	height 5:$z0$, class 1 because $z\geq0$

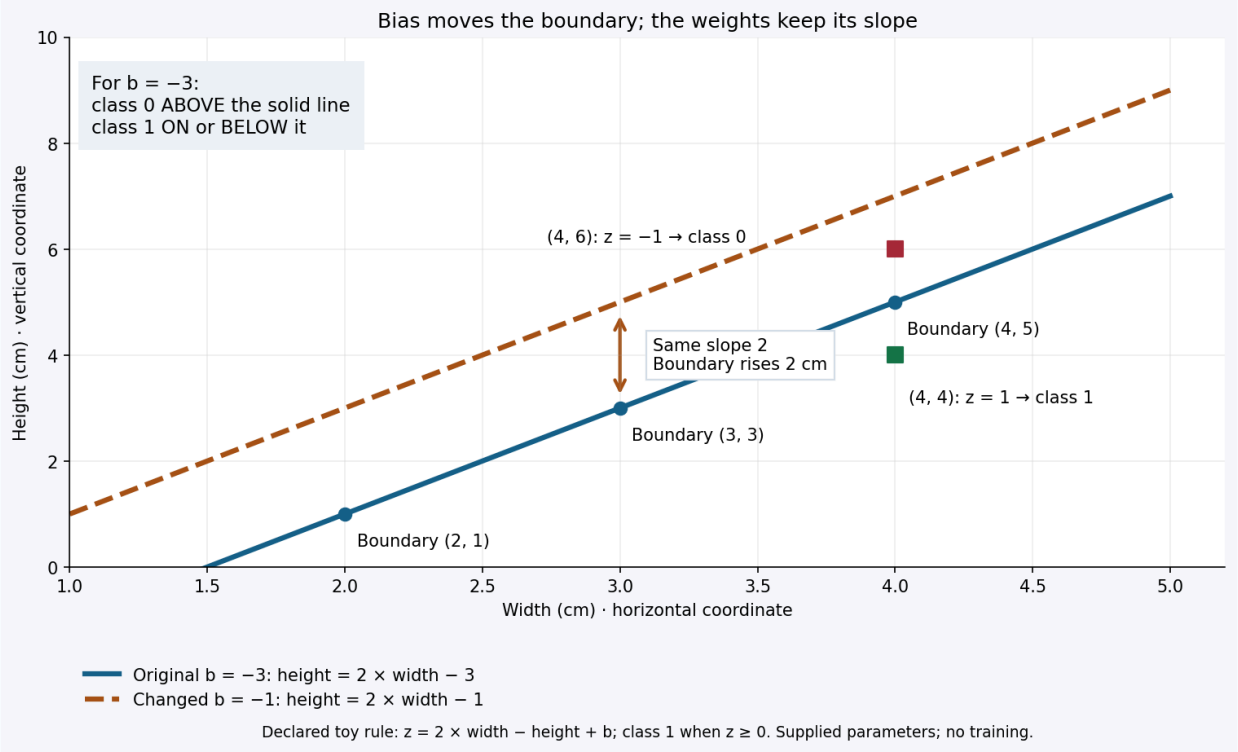
Set z to zero to locate the boundary: 2 times width - height - 3 = 0, so height = 2 times width - 3. Plot width on the horizontal axis and height on the vertical axis. Boundary points include (2, 1), (3, 3) and (4, 5).

Join them with a straight line and label the axes in cm.

At width 3, height 2 gives score 1 and class 1; height 3 gives score 0 and class 1; height 4 gives score -1 and class 0. Under this axis convention class 1 lies on or below the boundary.

One point on each side checks that orientation.

Labelled original and shifted boundary



Runnable score/class and numerical assumptions

```
def classify(width, height, bias=-3):
    score = 2 * width - height + bias
    return score, int(score >= 0)
```

```
for height in [2, 3, 4]:
    print(classify(3, height))
# (1, 1), (0, 1), (-1, 0)
```

The code assumes valid numerical measurements in the agreed units. Changing >= to > changes the equality case, even though the drawn line remains in the same place.

Connect the visual and calculation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change only the bias and keep the weights fixed

> Predict how much each score changes before checking.

Current boundary task and response

- For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the pictured bias change

- Change: Change bias from -3 to-1.

▼ Check after predicting

Prediction: A score becomes 2 and B score 0; both predict class 1.

Reason: Increasing bias by2 adds 2 to every score without changing the weights.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the same offset through two inputs

> Every score rises by the same amount even when only one prediction changes.

Complete picture facts and formula contract

Move a toy boundary with a bias while keeping its **input** weights fixed.

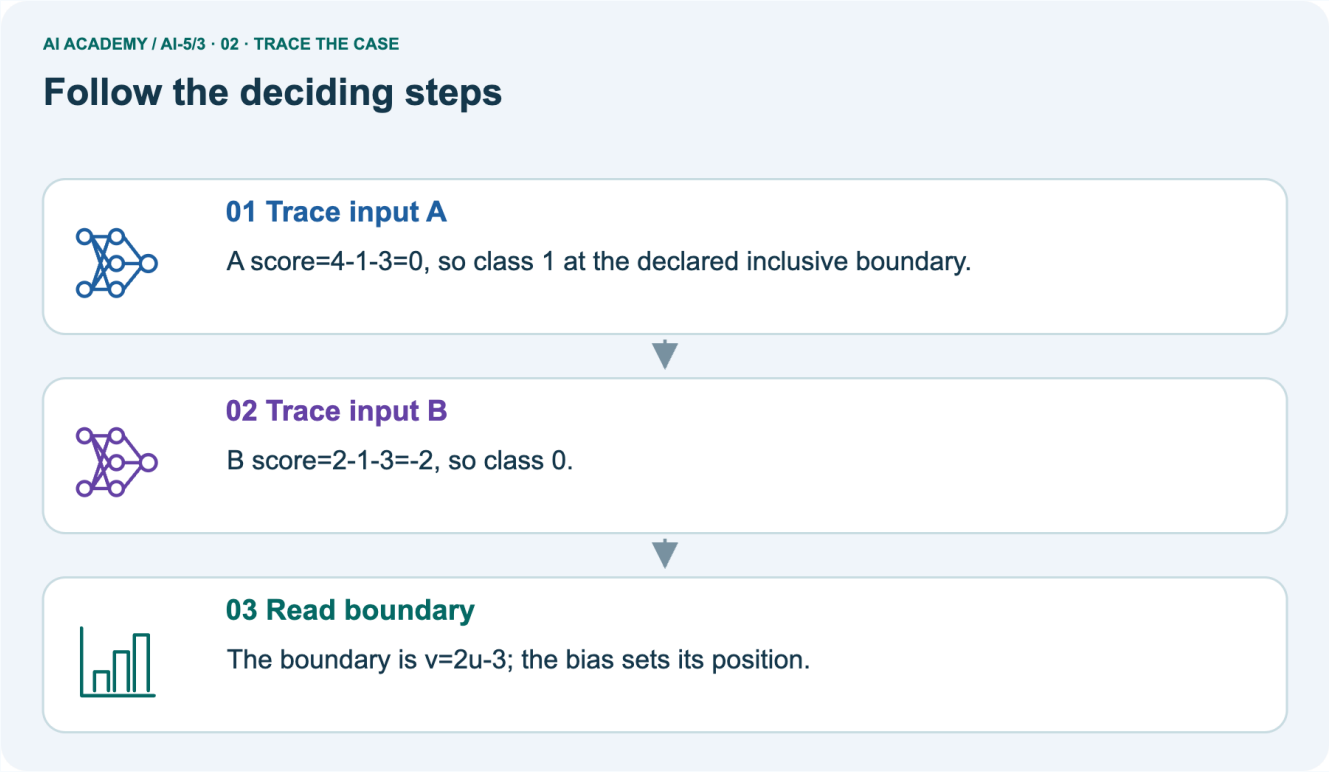
Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses the same formula z=2u−v+b with b=−3. A[2, 1]scores 0, class 1 at equality; B[1, 1]scores−2, class 0. Change only b to−1:both scores rise 2, A→2 and B→0, bothclass 1. The boundary changes from v=2u−3 to v=2u−1; its slope stays 2. The plotted parcel example uses width/height names for the same coordinate roles.

The hand-chosen shift is a different supplied model, not a learned improvement or probability estimate.

Trace original scores and included equality



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Trace input A	A score=4-1-3=0, so class 1 at the declared inclusive boundary.
2. Trace input B	B score=2-1-3=-2, so class 0.
3. Read boundary	The boundary is v=2u-3; the bias sets its position.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

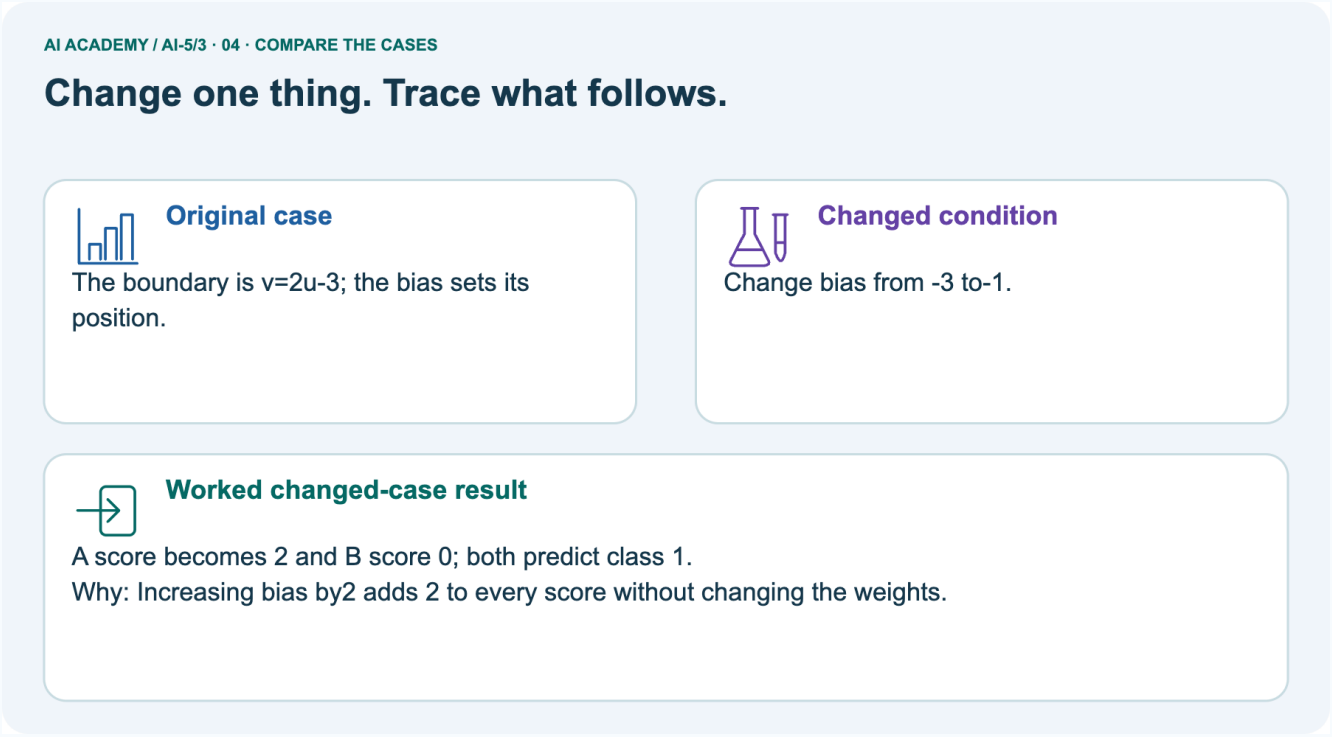
The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

Predict the changed bias

- Change one thing: Change bias from -3 to-1. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain limits and changed prediction

- A linear boundary is a toy classifier rule. Raw score size does not automatically describe confidence or probability.

- Read the labelled picture: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the separate parcel[3,4]change

Hold the weights fixed while shifting the boundary

BIAS	SCORE AT [3, 3]	BOUNDARY EQUATION
b is -3	6 - 3 - 3 gives 0	height = 2 width - 3
b is -1	6 - 3 - 1 gives 2	height = 2 width - 1
Increase b by 2	Every score rises by 2	Slope remains 2

The boundary moves parallel to itself because the weights stay fixed. An increased score does not always change a class: only inputs crossing the declared threshold change.

For [3, 4], compare z and prediction with bias -3 and bias -1. Explain which parameter changed.

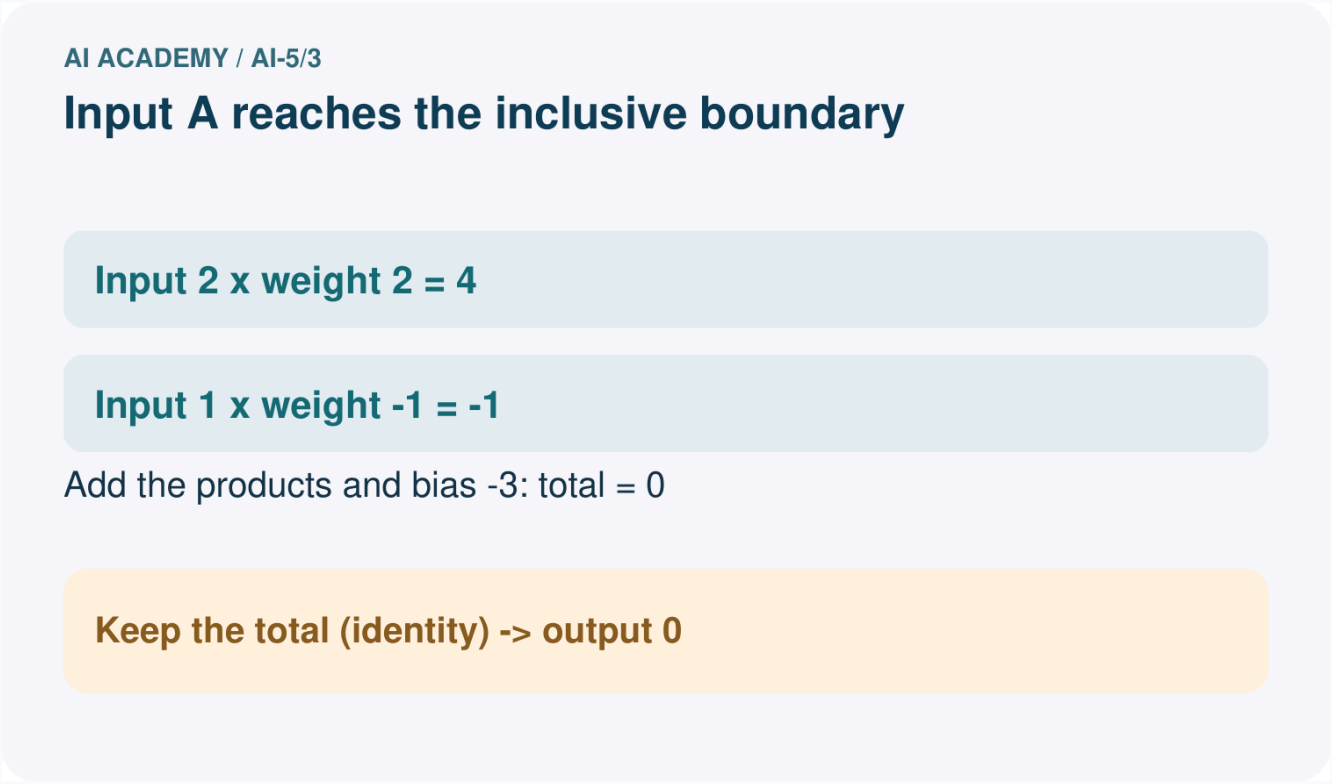
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Bias terms and decision boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/3

Bias terms and decision boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace input A

A score=4-1-3=0, so class 1 at the declared inclusive boundary.



2. Trace input B

B score=2-1-3=-2, so class 0.



3. Read boundary

The boundary is v=2u-3; the bias sets its position.

CHANGE ONE THING

Change bias from -3 to-1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair equality mistakes and automatic-improvement claims

> Moving a boundary does not by itself supply better labelled outcomes.

Origin,valid measurement domain and fixed slope

Without a bias, this score would be z = 2 times width - height and its zero line would pass through (0, 0). The offset lets the boundary move away from the origin.

In this parcel schema zero dimensions are invalid measurements; the origin is still useful as a point on the extended mathematical graph.

Changing bias from -3 to -1 increases every score by 2, but does not rotate the line. Changing a weight can rotate or otherwise change the boundary. Neither change is automatically an improvement.

To compare models, retain labels, a declared metric, the same evaluation records and the separate **training**, validation and test roles.

Test inclusive versus strict equality

- At [3, 3], compare **prediction** under z >= 0 with **prediction** under z > 0. Does changing this comparison move the zero line?

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Practise algebra,side checks and explanation

> Each learner shows a score and deciding comparison.

Current task and response

- Use z = 2 times width - height - 3 at [5, 4]. Give the weighted sum, bias, final score and **prediction** under z >= 0.

> **For [3, 4], compare z and prediction with bias -3 and bias -1. Explain which parameter changed.**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “**Bias term**s and decision boundaries”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Bias term**s and decision boundaries. Proposed explanation: “Add bias to each contribution”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured boundary reasoning

- Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the new independent model

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Plot and label the boundary

Check your reasoning

Explain the picture

Investigate both sides,equality and a controlled boundary shift

> The complete probe set checks mathematical meaning and input assumptions.

Complete current task and runnable classifier

For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

```
def classify(width, height, bias=-3):
    score = 2 * width - height + bias
    return score, int(score >= 0)

for height in [2, 3, 4]:
    print(classify(3, height))
# (1, 1), (0, 1), (-1, 0)
```

All supplied probes

Complete boundary probe	Expected score / class / action
Normal(4, 4), b−3	z1, class 1; below original line
Opposite side(4, 6), b−3	z−1, class 0; above original line
Equality(3, 3), b−3	z0, class 1 under≥0; class 0 if the rule changes to>0
One-factor change(3, 4), b−3→−1	z−1→1, class 0→1; only bias changes
Missing height(3, None)	Outside valid numeric-measurement contract; request height before calling
Shifted units:height 40 mm passed as40 cm	Formula yields−37 instead of−1 at width 3; correct units before scoring
Increase bias without labelled outcomes	Can compute new scores; cannot establish accuracy or improvement

Predict,plot,run or trace,and explain

1. Predict the complete probes, then use the supplied classify function in separate calls. Keep one output or error per call; mark paper traces as traces. On the graph mark the original boundary points and both sides before adding the dashed changed line. The function assumes valid numbers in cm and does not validate missing values or units; check the input schema first.

   For an actual model comparison, retain labels and the same evaluation cases and split roles. Changes informed by final errors create a development revision, not an untouched final result.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: derive a different model's boundary

> Use the new weights,bias and equality rule supplied here.

Complete new independent formula and tasks

A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Analyse a new model independently

Attempt this without hints

A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the formula,threshold and graph in one project record

> A reader should be able to recompute the same prediction.

Project artifact and evidence table

Add a labelled **bias term**s and decision boundaries **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a weighted score with a **bias term**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual evidence and scope

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a boundary point and its equality decision

> Explain the controlled change without claiming unmeasured improvement.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Calculate a weighted score with a **bias term**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Bias term	An additive parameter applied after a weighted sum
Decision boundary	Input locations where the declared classification rule changes class
Affine function	A weighted sum plus a constant offset
Threshold rule	A stated comparison that converts a score into a class

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Use z = 2 times width - height - 3 at [5, 4]. Give the weighted sum, bias, final score and **prediction** under z >= 0.
- At [3, 3], compare **prediction** under z >= 0 with **prediction** under z > 0. Does changing this comparison move the zero line?
- A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent weights[−1,2]model for the first individual attempt.

Current explanation and response

- The boundary states what this particular formula predicts. It does not establish the true classes of all possible **input**s.

  A point very far from the line need not have a reliable **prediction** when the model, features or data are inappropriate. The raw score is not calibrated confidence or a physical distance from the line.

- Next week, we transform scores with activation functions before discussing layers.

- A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

> **A team changes b after seeing final-test mistakes and reports the same cases as an untouched test of the new model. Explain the error and the next step.**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Hold the weights fixed while shifting the boundary

BIAS	SCORE AT [3, 3]	BOUNDARY EQUATION
b is -3	6 - 3 - 3 gives 0	height = 2 width - 3
b is -1	6 - 3 - 1 gives 2	height = 2 width - 1
Increase b by 2	Every score rises by 2	Slope remains 2

The boundary moves parallel to itself because the weights stay fixed. An increased score does not always change a class: only inputs crossing the declared threshold change.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Move a toy boundary with a bias while keeping its **input** weights fixed.

By the end, I can	How I will show it
Calculate a weighted score with a bias term	Explain the deciding step, show a trace or test, and state one limit.
Locate and test a two-feature decision boundary	Explain the deciding step, show a trace or test, and state one limit.
Explain how changing the bias shifts a boundary without training	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Move a toy boundary with a bias while keeping its **input** weights fixed.
- Which labelled step matches this detail? The boundary is v=2u-3; the bias sets its position.
- 1. Trace **input** A
- 2. Read boundary
- 3. Trace **input** B
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Read boundary The boundary is v=2u-3; the bias sets its position. The bias adds the same offset after the **input** contributions.

It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

Changed-case question and optional hint

- Changed-case practice: Change bias from -3 to-1. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. A score becomes 2 and B score 0; both predict class 1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A score becomes 2 and B score 0; both predict class 1. Increasing bias by2 adds 2 to every score without changing the weights.

Repair and retain the record

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Bias terms and decision boundaries: the case

AI ACADEMY / AI-5/3 · 01 · LOOK AT THE FACTS

Bias terms and decision boundaries: the case

The situation

Move a toy boundary with a bias while keeping its input weights fixed.

Supplied evidence

Score z=2u-v-3; predict class 1 when z>=0.
InputsA=[2, 1], B=[1, 1].
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain bias terms and decision boundaries using a traced example and a stated limitation.

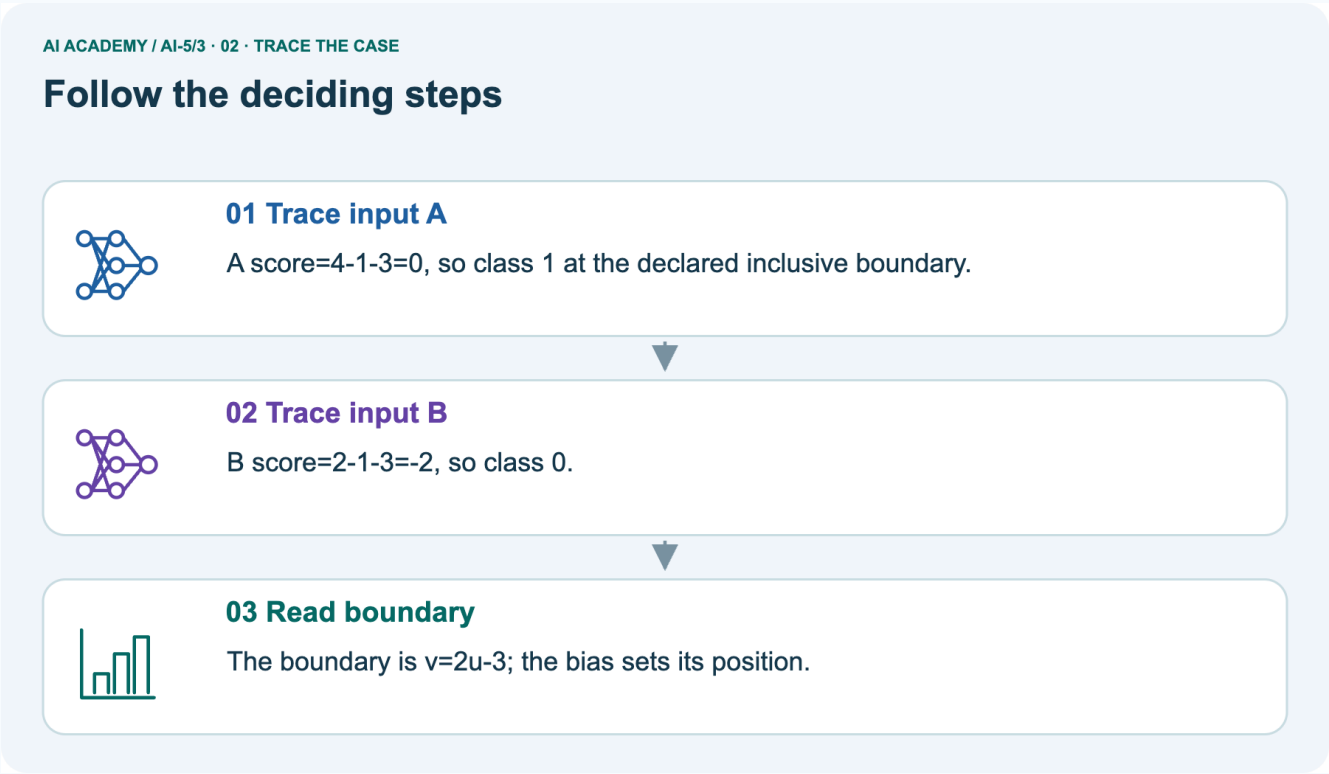
Follow the labelled picture with the matching explanation. Open the vector diagram

Move a toy boundary with a bias while keeping its **input** weights fixed. Supplied facts: Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

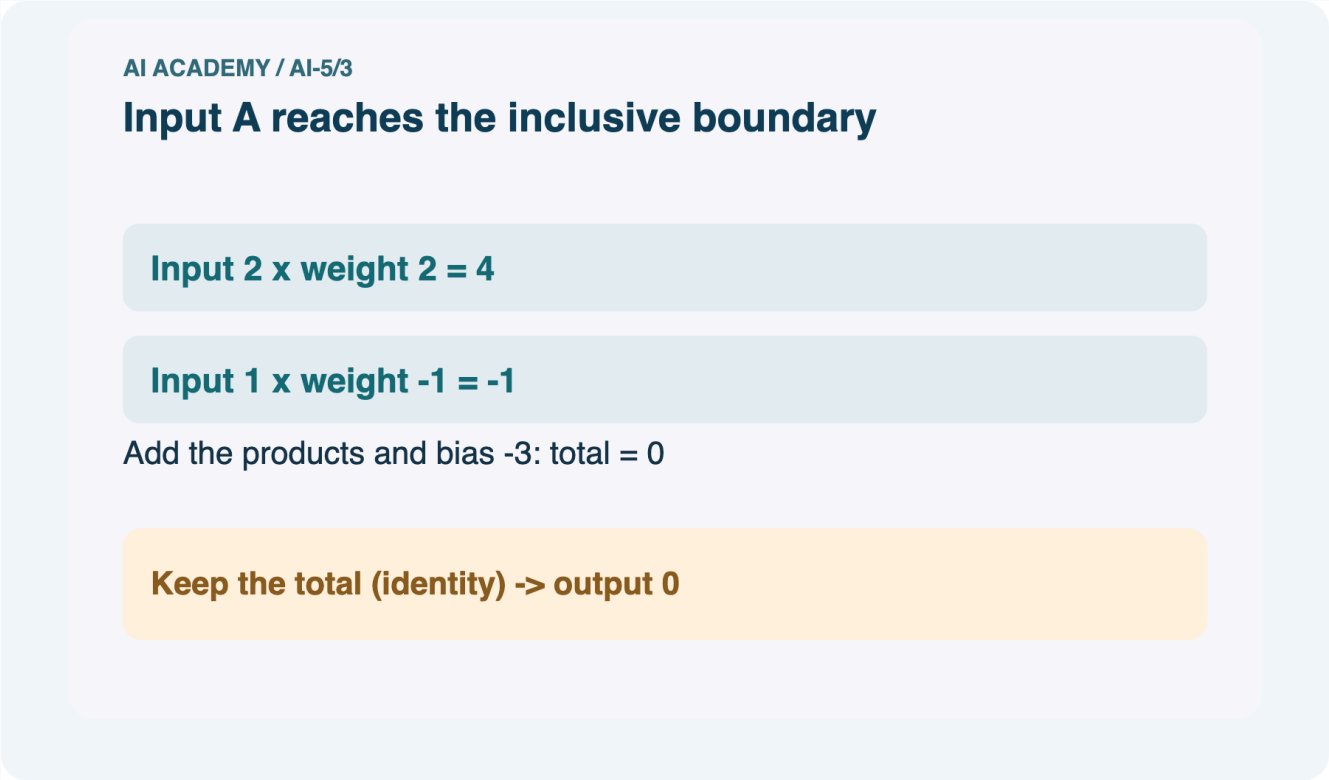


Follow the labelled picture with the matching explanation. Open the vector diagram

Trace **input** A: A score=4-1-3=0, so class 1 at the declared inclusive boundary. → Trace **input** B: B score=2-1-3=-2, so class 0. → Read boundary: The boundary is v=2u-3; the bias sets its position.

Point to each numbered stage and explain which recorded **input** it uses.

03. Input A reaches the inclusive boundary

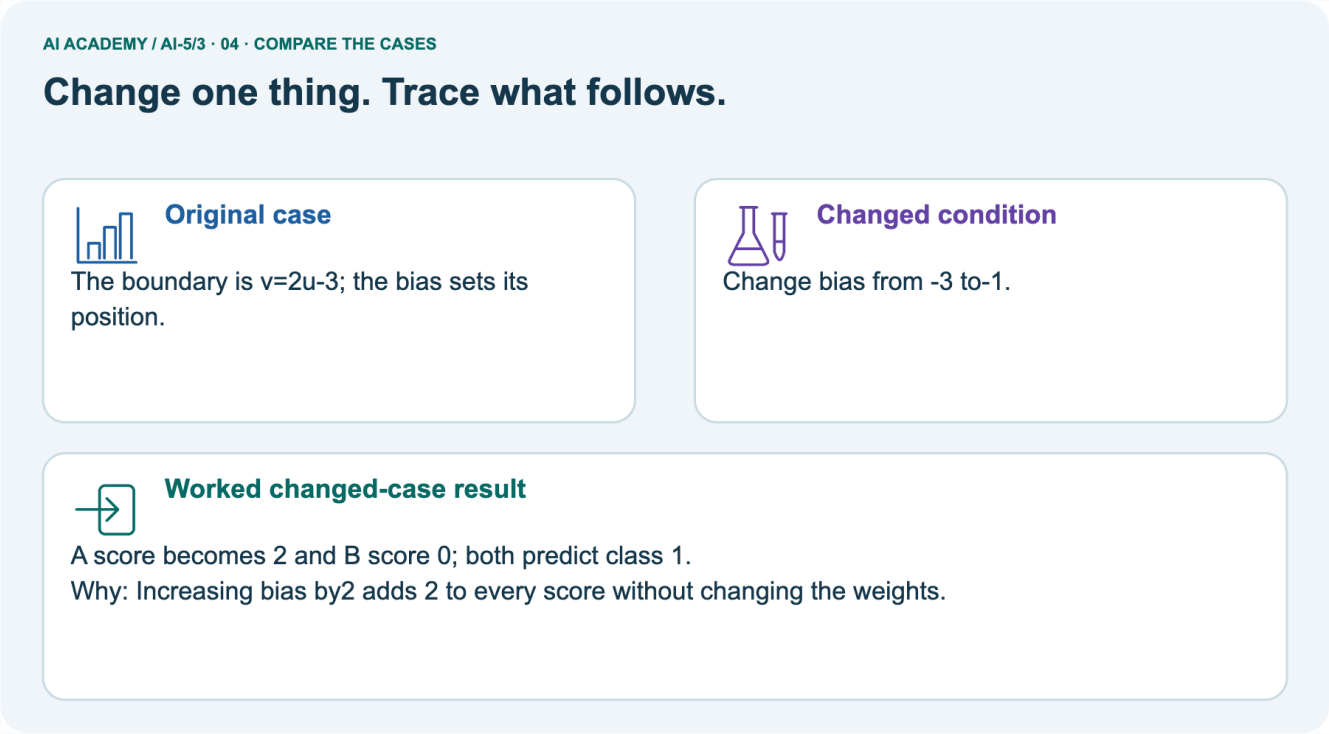


Follow the labelled picture with the matching explanation. Open the vector diagram

Input A reaches the inclusive boundary. **Input**s: [2, 1]; weights: [2, -1]; bias: -3; activation: identity; **output**: 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



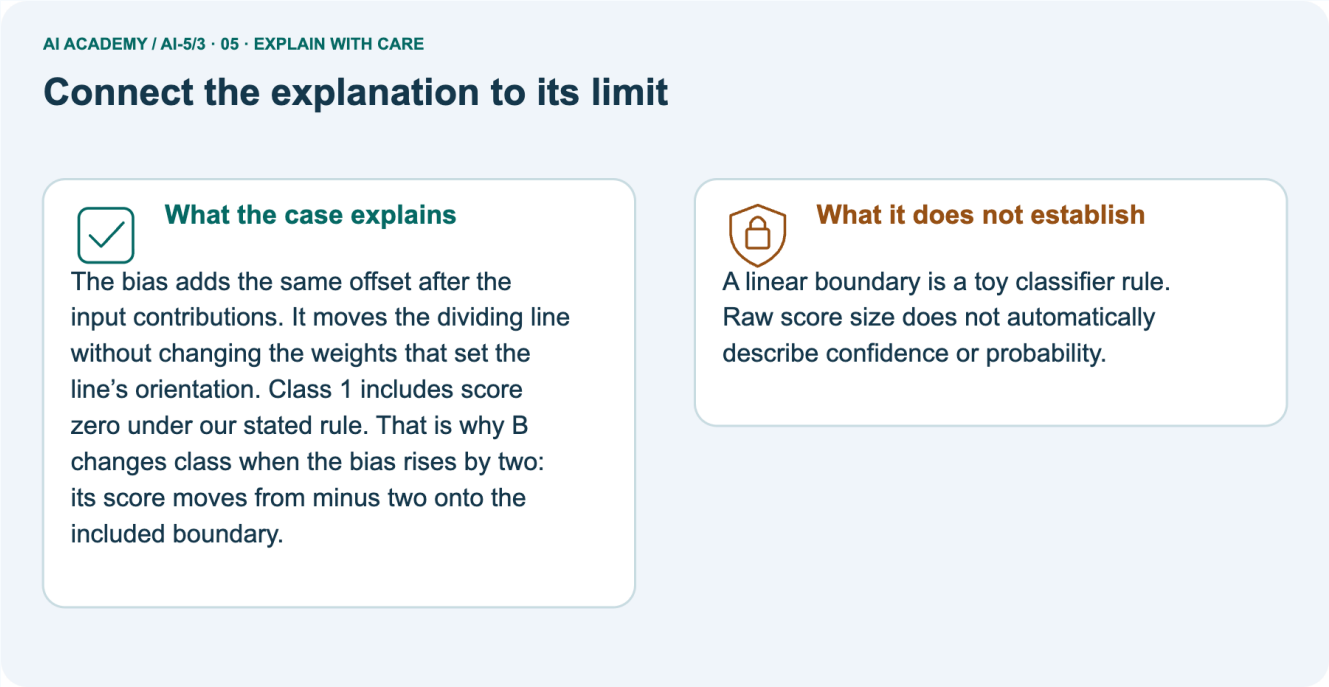
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The boundary is v=2u-3; the bias sets its position. Changed condition: Change bias from -3 to-1. Expected result: A score becomes 2 and B score 0; both predict class 1.

Reason: Increasing bias by2 adds 2 to every score without changing the weights.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

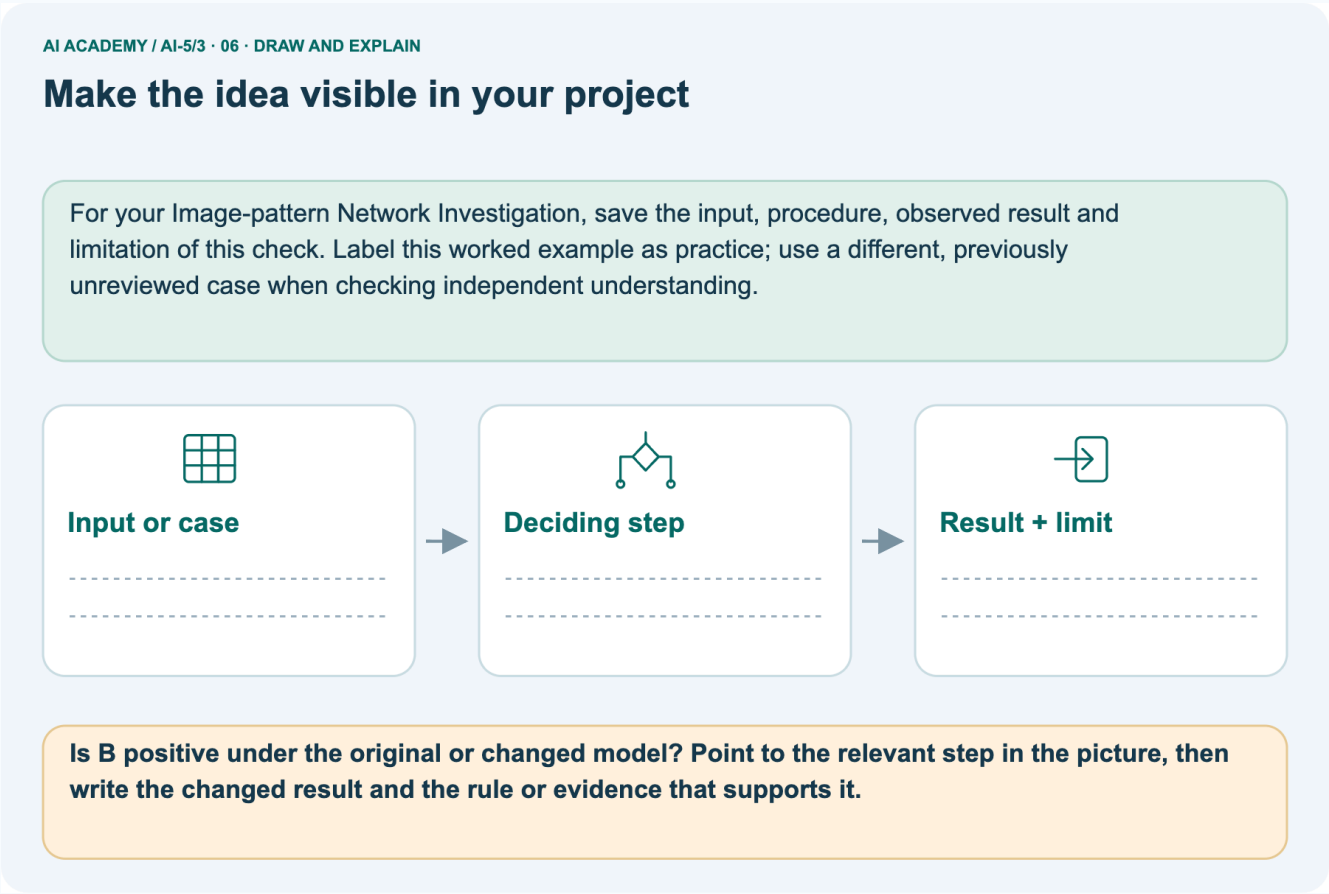
The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary. Limit: A linear boundary is a toy classifier rule.

Raw score size does not automatically describe confidence or probability.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Bias terms and decision boundaries: the case

02. Follow the deciding steps

03. Input A reaches the inclusive boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.